

GIT16

:tada:

Initial commit or project bootstrap.

:bug:

Fix functional bugs in RTL, verification, Python, scripts, or flow code.

:ambulance:

Fix urgent breakage: failing regression, broken CI, lab blocker, release blocker, or tapeout blocker.

:sparkles:

Add a complete new feature.

:construction:

Commit incomplete work intentionally.

:boom:

Breaking interface, register map, protocol, API, timing contract, or CLI change.

:recycle:

Refactor without intended behavior change.

:art:

Improve structure, formatting, naming, or readability without behavior change.

:fire:

Remove obsolete code, files, stale scripts, or generated artifacts.

:coffin:

Remove dead signals, unreachable FSM states, unused functions, or obsolete helpers.

:wastebasket:

Deprecate code or behavior that should be removed later.

:truck:

Move or rename files, modules, packages, paths, or resources.

:rewind:

Revert a previous change.

:twisted_rightwards_arrows:

Merge branches. Do not use for muxing, routing, or arbitration.

:see_no_evil:

Add or update `.gitignore`.

:bookmark:

Release or version tag.

CI2

:construction_worker:

CI pipelines, GitHub Actions, build matrix, scheduled jobs, or regression automation.

:green_heart:

Fix broken CI, regression automation, build matrix, or tool-cache issue.

RTL12

:bridge_at_night:

Clock-domain crossing, async boundary, synchronizer, handshake bridge, or reset-domain crossing.

:rightwards_pushing_hand:

Valid/ready pushback, stall propagation, skid buffers, flow control, or downstream backpressure.

:electric_plug:

Wire ports, buses, protocols, register interfaces, AXI/APB/TileLink/Wishbone connections, or top-level integration.

:abacus:

Arithmetic, bit-width, fixed-point, saturation, rounding, accumulator, multiplier, divider, or datapath logic.

:vertical_traffic_light:

Arbitration, priority, scheduling, traffic shaping, QoS, credits, or grant logic.

:compass:

Address maps, memory maps, topology, routing tables, NoC coordinates, or integration direction.

:dna:

Generics, parameters, SKUs, feature variants, configurable widths/depths, or generated RTL variants.

:fire_extinguisher:

Fix X-prop, glitch risk, unsafe reset behavior, metastability risk, race, or silicon-escape-class hazard.

:magnet:

Board-level work, IO, pins, PHYs, lab-only fixes, FPGA bring-up, or constraints tied to real hardware.

:satellite_antenna:

PCIe, Ethernet, DDR, SerDes, SPI, I2C, UART, or other external protocol/link behavior.

:safety_pin:

Small localized hardware workaround or mitigation for a known issue.

:screwdriver:

Small mechanical RTL adjustment too narrow for a bigger category.

SCRIPTS2

:hammer:

Build, sim, synth, lint, formal, regression, packaging, or helper scripts.

:wrench:

Tool config, environment config, CLI config, YAML/TOML/JSON config, synthesis/sim flags, or parameters.

VERIFICATION7

:white_check_mark:

Add, update, or fix passing tests, regressions, assertions, formal checks, or checkers.

:test_tube:

Add a failing test or reproducer for a known bug.

:rotating_light:

Fix compiler, linter, CDC, synthesis, simulation, or timing warnings.

:game_die:

Randomized testing, constrained-random stimulus, fuzzing, seeds, or randomized regressions.

:microscope:

Add probes, traces, counters, waveform hooks, debug CSRs, or observability.

:safety_vest:

Add validation, guards, illegal-state checks, protocol checks, or defensive checks.

:seedling:

Add seed files, sample inputs, golden data, or deterministic regression data.

PPA4

:zap:

Improve speed, throughput, latency, benchmark result, or runtime.

:chart_with_downwards_trend:

Reduce LUTs, FFs, gates, SRAM, BRAM, DSPs, cells, or general resource usage.

:chart_with_upwards_trend:

Accept area, power, or resource increase intentionally for feature, timing, verification, or performance.

:battery:

Clock gating, power gating, idle behavior, low-power mode, activity reduction, or power optimization.

DOCS4

:memo:

Add or update docs, specs, register maps, integration notes, or README content.

:bulb:

Add or update source comments explaining hardware behavior, assumptions, or constraints.

:pencil2:

Fix typos in docs, comments, specs, or messages.

:page_facing_up:

Add or update license.

DEPENDENCIES6

:arrow_up:

Upgrade dependency, vendor IP, tool version, package, or submodule.

:arrow_down:

Downgrade dependency, vendor IP, tool version, package, or submodule.

:pushpin:

Pin exact versions of tools, IPs, packages, lockfiles, containers, or submodules.

:heavy_plus_sign:

Add a dependency, package, included IP, submodule, or tool requirement.

:heavy_minus_sign:

Remove a dependency, package, included IP, submodule, or tool requirement.

:alien:

Update code due to external API, protocol, vendor IP, tool output, or report-format changes.

PYTHON5

:snake:

Python-specific implementation work when no more precise standard Gitmoji applies.

:label:

Add or update Python type hints, dataclasses, TypedDicts, Pydantic models, SV typedefs, enums, or structs.

:goal_net:

Improve exceptions, parser errors, failure handling, or user-facing error messages.

:monocle_face:

Data exploration, report inspection, one-off analysis, or investigation scripts.

:thread:

Threading, multiprocessing, async, job queues, or concurrent report processing.

TIMING2

:scissors:

Reduce combinational depth, retime logic, split paths, or improve Fmax.

:stopwatch:

Benchmark instrumentation, cycle counting, latency measurement, or timing measurement infrastructure.

INFRASTRUCTURE2

:bricks:

Hardware-flow infrastructure: runners, artifact storage, caches, job orchestration, or shared tool setup.

:package:

Add or update generated bundles, packaged IP, release artifacts, reports, or bitstreams.